

UST-BARGUZIN, Russian Federation

WMO: 306350

Lat: 53.417N

Lon: 109.017E

Elev: 1511

StdP: 13.91

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-22.9	-31.8	1.0	-27.5	-27.3	1.3	-22.5	15.0	5.9	13.1	2.0	2.6	90	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	79.0	63.7	74.9	62.3	71.1	61.2	66.1	74.8	64.1	71.5	62.4	69.2	5.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	91.0	69.4	61.3	85.7	67.6	59.7	80.9	65.9	31.8	74.6	30.2	71.6	29.0	69.3	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.2	9.7	-34.4	87.8	6.4	3.4	-38.9	90.2	-42.7	92.2	-46.2	94.1	-50.8	96.6		
(5)			-33.8	69.3	5.4	2.2	-37.7	70.9	-40.9	72.2	-43.9	73.4	-47.9	75.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.0	-4.4	-1.1	12.8	30.6	41.6	53.7	61.4	59.3	47.8	33.9	17.5	4.5		
	DBStd	24.14	10.54	11.32	11.57	7.29	5.72	6.24	5.55	5.38	6.39	7.12	10.48	10.94		
	HDD50	8158	1688	1431	1155	582	268	29	0	3	116	500	975	1411		
	HDD65	12839	2153	1851	1620	1032	725	344	142	191	515	965	1425	1876		
	CDD50	844	0	0	0	0	9	140	353	290	51	1	0	0		
	CDD65	49	0	0	0	0	0	4	31	14	0	0	0	0		
	CDH74	441	0	0	0	0	0	64	258	115	4	0	0	0		
	CDH80	83	0	0	0	0	0	8	57	18	0	0	0	0		
	WSAvg	4.4	4.8	3.8	3.8	4.3	4.4	4.0	3.9	3.9	4.4	4.4	5.5	6.0		
	PrecAvg	14.6	0.5	0.2	0.2	0.5	0.7	1.3	3.1	2.6	1.6	1.0	1.4	1.5		
(15) Precipitation	PrecMax	19.4	2.0	0.7	0.9	1.6	1.7	4.6	9.0	5.1	3.9	2.2	2.7	2.4		
	PrecMin	10.8	0.2	0.0	0.0	0.1	0.1	0.0	0.4	0.5	0.6	0.1	0.4	0.5		
	PrecStd	2.4	0.4	0.2	0.2	0.4	0.4	1.1	2.5	1.1	0.7	0.6	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	30.5	43.2	55.6	68.0	80.8	86.0	82.3	71.9	55.8	43.5	34.6		
		MCWB	24.5	26.1	34.9	42.4	51.9	61.7	66.0	64.4	58.1	47.2	38.2	30.4		
	2%	DB	21.7	24.7	37.5	48.7	60.7	75.0	80.5	77.0	66.4	51.2	39.8	29.3		
		MCWB	19.3	21.9	31.8	39.6	48.3	59.6	64.9	63.6	56.6	44.8	35.4	26.0		
	5%	DB	16.0	20.6	33.5	44.7	56.8	70.1	76.3	72.7	62.3	47.8	35.9	25.7		
		MCWB	14.2	18.3	29.0	37.3	46.3	58.2	64.1	61.9	54.5	42.3	31.7	22.8		
	10%	DB	10.7	16.4	30.1	41.5	53.2	65.9	72.3	69.2	59.0	44.9	32.4	21.2		
		MCWB	9.5	14.6	26.5	35.4	44.3	56.4	62.7	60.7	52.4	39.9	28.9	18.8		
	0.4%	WB	25.3	26.5	35.8	44.0	52.9	64.6	69.6	68.5	61.0	49.1	38.9	30.5		
		MCDB	27.4	30.3	41.6	54.1	64.3	74.9	79.1	77.5	67.9	54.5	43.6	34.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	19.5	22.0	32.2	40.4	49.5	61.7	67.2	64.8	57.9	45.6	35.8	25.9		
		MCDB	21.8	24.6	37.4	47.5	58.5	71.0	76.5	72.6	64.7	50.0	39.6	29.0		
	5%	WB	14.2	18.3	29.3	37.8	47.1	59.4	65.4	63.2	55.6	42.9	32.0	22.9		
		MCDB	16.0	20.5	33.3	43.6	55.2	68.0	73.3	70.2	60.8	47.4	35.4	25.8		
	10%	WB	9.5	14.6	26.5	35.9	44.9	57.3	63.6	61.7	53.3	40.3	29.0	18.9		
		MCDB	10.7	16.4	30.1	41.0	52.1	64.7	70.5	67.9	58.3	44.5	32.4	21.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	20.0	20.6	16.0	17.5	16.9	15.7	16.1	16.2	13.8	11.5	11.8		
		MCDBR	17.6	20.5	18.8	20.7	25.1	26.4	24.0	23.2	21.8	18.7	15.0	16.2		
		MCWBR	16.0	17.9	13.7	13.1	14.5	13.4	11.1	12.2	13.2	13.5	12.1	14.2		
		MCDBR	17.6	20.1	18.1	19.0	22.4	21.6	19.4	19.3	18.6	17.7	14.8	16.3		
	5% WB	MCWBR	16.1	17.7	13.5	12.7	13.8	12.5	10.2	11.1	11.9	13.4	12.4	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.294	0.322	0.371	0.431	0.448	0.430	0.391	0.343	0.312	0.281	0.261		
	taud		2.076	2.056	2.085	2.183	2.113	2.174	2.282	2.370	2.453	2.396	2.140	2.035		
	Ebn at Noon		218	249	269	271	259	256	259	263	265	247	209	189		
	Edh at Noon		23	32	39	41	46	44	39	34	28	23	21	19		
(44) All-Sky Solar Radiation	RadAvg		312	624	1074	1500	1813	1942	1771	1481	1066	610	297	203		
	RadStd		26	38	42	98	112	116	148	66	87	61	31	17		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-3.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	N/A	-3.28	-4.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page